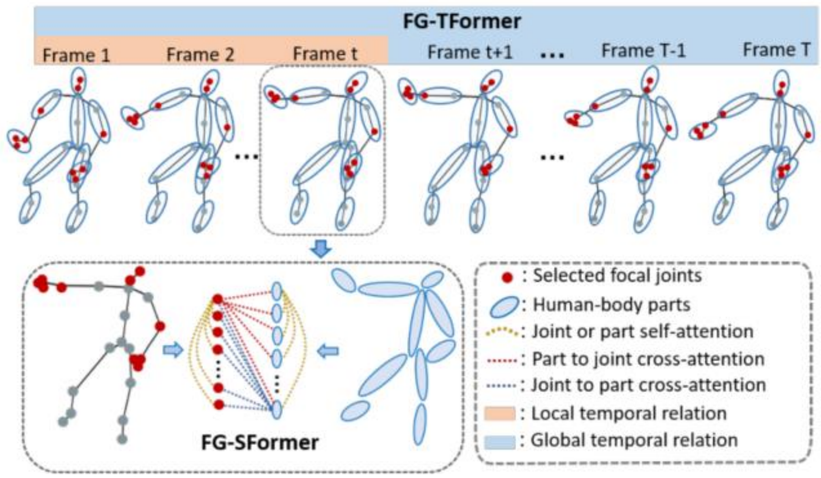
Focal and Global Spatial-Temporal Transformer for Skeleton-based Action Recognition

Zhimin Gao , Peitao Wang , Pei Lv , Xiaoheng Jiang , Qidong Liu , Pichao Wang, Mingliang Xu, Wanqing Li

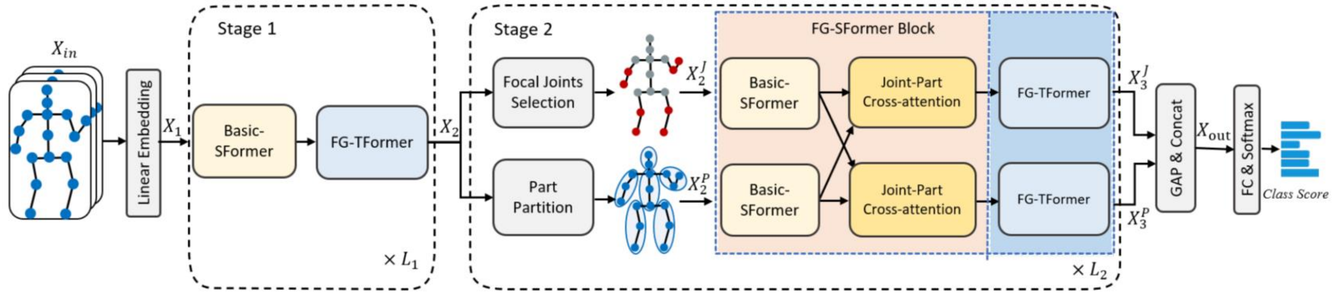
ACCV 2022 Macau

Motivation

- Despite the great success achieved by transformer in various vision tasks, it is still underexplored for skeleton-based action recognition with only a few attempts.
- Those transformer based methods calculate the self-attention equally for all the joints in both the spatial and temporal dimensions, undervaluing the effect of discriminative local joints and the short-range temporal dynamics.
- One action can be distinguished by the co-movement of: (1) some critical local joints, (2) global body parts, and (or) (3) joint-part interaction.

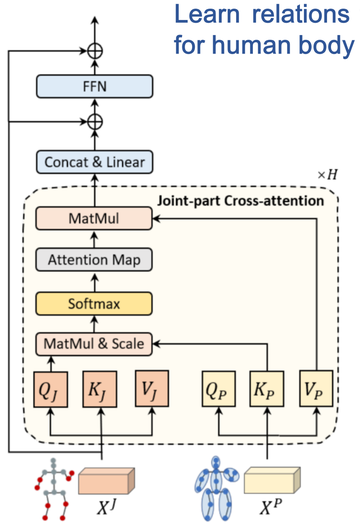


Framework Overview: FG-STFormer



FG-SFormer

Learn relations for adaptively identified focal joints and for human body parts, as well as their interactions.



- Focal Joints Selection

$$S = \mathrm{sigmoid}(X_2 W_p / \|W_p\|)$$
$$\mathrm{idx} = \mathrm{sort}(S, K)\ ,$$
$$X_2^J = X_2(\mathrm{idx}, :, :)$$

- Global-level Part Partition Encoding

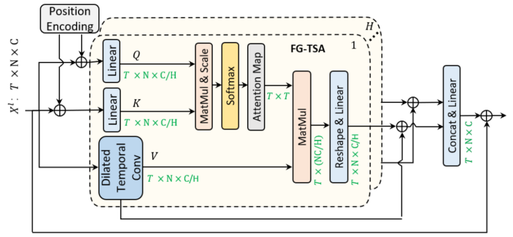
  Concatenate the joints in the same body part and transform them into one part feature embedding via a linear layer.

- Joint-Part Cross-attention:

$$\mathrm{Attention}(Q_J, K_P, V_P) = \mathrm{softmax}(\frac{Q_J {K_P}^T}{\sqrt{d}})V_P$$
$$= A_{jp} V_P$$

FG-TFormer

Enable transformer to focus on both the important local and the global temporal relations of joints explicitly.



- Dilated temporal convolution is adopted to generate V:

$$Q_h^l,\ K_h^l \in \mathbb{R}^{N \times T \times \frac{C}{H}} \qquad V_h^l = \mathrm{TCN}_{\mathrm{dilate}}(X^l)_h \in \mathbb{R}^{N \times T \times \frac{C}{H}}$$

- One FG-TFormer block is formulated as:

$$X^{l+1} = \mathrm{Concat}[\mathrm{head}(X^l)_1, ..., \mathrm{head}(X^l)_H] W_O + X^l$$
$$\mathrm{head}(X^l)_h = [\mathrm{Attention}(Q_h^l, K_h^l)\mathrm{TCN}_{\mathrm{dilate}}(X^l)_h] W_h + \mathrm{TCN}_{\mathrm{dilate}}(X^l)_h$$

Experimental Results

- **Ablation study**

Components in FG-SFormer block

Methods	Focal Joints Selection	Part Branch	JP-CA	Acc (%)
Basic-SFormer	-	-	-	87.8
A	√	-	-	88.3
B	√	√	-	89.1
C	√	√	√	**89.5**

Network configurations

Methods	Stage 1 L_1	Stage 2 L_2	Acc (%)
A	4	4	88.8
B	5	3	89.0
C	6	2	**89.5**
D	7	1	89.0

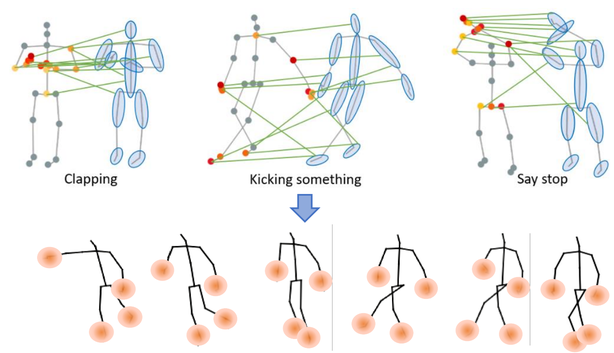
Number of focal joints

K	3	6	9	12	15	18	21	25
Acc (%)	88.6	88.9	89.0	89.2	**89.5**	89.2	89.3	89.1

Components in FG-TFormer block

Methods	Global MSA	TCN_{dilate}	Acc (%)
Basic-TFormer	√	×	82.8
FG-TFormer	×	√	89.1
	√	√	**89.5**

- **Selected focal joints and learned joint-part interactions of some actions**



- **Results on NTU-60 and NTU-120 datasets**

Methods	Year	NTU-60 X-Sub (%)	NTU-60 X-View (%)	NTU-120 X-Sub (%)	NTU-120 X-Set (%)
GCN-based Methods					
ST-GCN [6]	2018	81.5	88.3	70.7	73.2
2s-AGCN [37]	2019	88.5	95.1	82.9	84.9
DGNN [52]	2019	89.9	96.1	-	-
Shift-GCN [49]	2020	90.7	96.5	85.9	87.6
PA-ResGCN-B19 [53]	2020	90.9	96.0	87.3	88.3
Dynamic GCN [38]	2020	91.5	96.0	87.3	88.6
MS-G3D [7]	2020	91.5	96.2	86.9	88.4
MST-GCN [54]	2021	91.5	96.6	87.5	88.8
CTR-GCN [40]	2021	92.4	96.8	88.9	**90.6**
STF [51]	2022	92.5	**96.9**	88.9	90.0
Transformer-based Methods					
DSTA [14]	2020	91.5	96.4	86.6	89.0
ST-TR [15]	2021	89.9	96.1	82.7	84.7
UNIK [55]	2021	86.8	94.4	80.8	86.5
STST [16]	2021	91.9	96.8	-	-
FG-STFormer (ours)	2022	**92.6**	96.7	**89.0**	**90.6**